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(54) **DEPOSING INITIARY COMPOSITIONS**

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ABSTRACT

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There is provided a composition and method of depositing an initiatory composition, said composition, comprising a:

(i) a nanothermite suspension of a metal (M) oxide and a metal (M') in a solvent, wherein the average particle size of the metal (M) oxide and a metal (M') is less than 1000 nm, provided that (M)≠(M'),

(ii) wherein said nanothermite suspension comprises a charging reagent comprising a reagent capable of forming a stable complex with each of the metal (M) oxide and the metal (M'), to form a metal (M) oxide complex, and a metal (M') complex that have the same electrostatic charge, such that said metal (M) oxide complex and a metal (M') complex repel each other in said suspension, wherein the admixture of the binder, nanothermite suspension charging reagent, has been caused to be mixed under Resonant Acoustic Mixing to provide a stable suspension of a nanothermite complex.

DEPOSING INITIARY COMPOSITIONS

DEPOSING INITIARY COMPOSITIONS

[0001] This application is a continuation of U.S. National Phase application Ser. No. 17/289,434, filed Apr. 28, 2021, and entitled “DEPOSING INITIARY COMPOSITIONS”, which claims priority to PCT/GB2019/053063, filed Oct. 30, 2019, which in turn claims priority to GB 1817918.4, filed Nov. 2, 2018, the entire disclosure of each of which is hereby incorporated by reference herein.

[0002] There is provided a composition and method of depositing initiatory compositions, specifically for filling small devices such as caps, and ignitors.

[0003] There is a desire to move away from typical sensitive initiatory compositions, and specifically those that require the use of lead based compositions.

[0004] Primers for medium and small calibre munitions are typically modest-sized components requiring no more than a few hundred milligrams of primary explosive or other initiating composition. Consequently, filling such components by hand requires significant dexterity. Filling processes involving dry powders are also prone to initiation via electrostatic discharge, which promotes safety concerns with regards to the operator.

[0005] According to a first aspect of present invention there is provided a nanothermite initiatory composition, comprising a:

[0006] (i) a nanothermite suspension of a metal (M) oxide and a metal (M') in a solvent, wherein the average particle size of the metal (M) oxide and a metal (M') is less than 1000 nm, provided that (M)≠(M'),

[0007] (ii) wherein said nanothermite suspension comprises a charging reagent comprising a reagent capable of forming a stable complex with each of the metal (M) oxide and the metal (M'), to form a metal (M) oxide complex, and a metal (M') complex, such that said metal (M) oxide complex and said metal (M') complex repel each other in said suspension,

[0008] wherein the admixture of the binder, nanothermite suspension charging reagent, has been caused to be mixed under Resonant Acoustic Mixing to provide a stable suspension of a nanothermite complex; in a highly preferred arrangement the nanothermite suspension further comprises a non-ionic surfactant.

[0009] The repelling of said metal (M) oxide complex and a metal (M') complex is caused by the charging reagent having the same electrostatic charge or providing steric hindrance. The effect will be determined by the choice of the charging reagent.

[0010] The nanothermite may be present in a stoichiometric ratio or fuel-rich ratio of fuel and oxidizer nanopowders. Stoichiometric ratio is defined as Equivalence Ratio (ER or ϕ)=1.0. Fuel-rich ratios, for example, may be ER (ϕ)=1.2.

$$\phi = \frac{\left(\frac{\text{Fuel}}{\text{Oxidizer}} \right)_{\text{Experimental}}}{\left(\frac{\text{Fuel}}{\text{Oxidizer}} \right)_{\text{Stoichiometric}}}$$

[0011] In a highly preferred arrangement the solvent may comprise a binder, in the range of from 0.1 to 16% w/w.

[0012] The binder may be any typically used binder in energetic material compositions. The binder may be selected

from a non-energetic binder and/or an energetic binder, present in the range of from 0.1% to 16% wt, more preferably in the range of 0.5 to 5% wt.

[0013] The binder may be a mixture of an energetic and non-energetic binder. Examples of suitable non-energetic binder materials are ethylene-vinyl acetate, esters, ego cellulose acetate, cellulose acetate butyrate, polyurethanes, polyesters, polybutadienes, polyethylenes, polyvinyl acetate and blends and/or copolymers thereof, or fluorinated binders. The fluorinated binders may be mono- or per-fluorinated, such as, for example PTFE, Viton etc.

[0014] Examples of suitable energetic binder materials which may be used alongside a non-energetic binder, may be nitrocellulose, polyvinyl nitrate, nitroethylene, nitroallyl acetate, nitroethyl acrylate, nitroethylmethacrylate, trinitroethyl acrylate, di nitropropyl acrylate, C-nitro polystyrene and its derivatives, polyurethanes with aliphatic C- and N-nitro groups, polyesters made from dinitrocarboxylic acids and dinitrodiol and homopolymers of 3-10 nitro-3 methyl oxetane (PolyNIMMO).

[0015] The nanothermite initiatory composition may be deposited directly into a device, or deposited to provide a loose composition that can be filled using conventional techniques, such as granules, rods, pellets, etc. The device to be filled may be any munition or subsystem, typically this may be a primer cup, initiator, or part of the explosive train.

[0016] In a preferred arrangement the nanothermite suspension may be caused to be evaporated, to provide a powdered nanothermite encapsulated in the binder material. The evaporation stage may be afforded in once in a filled device.

[0017] The admixture, comprising, for example, nanothermite suspension, binder, adhesive, optionally any curative, may be directly applied to any substrate or device by means of either extrusion, deposition or spray process, e.g. aerosol, nozzle printing or vapour deposition.

[0018] Plasticisers, such as non-energetic, energetic plasticisers or a combination therefore may be used.

[0019] The nanothermite solids loading in the suspension may be greater than 10% w/w, preferably of from 10% to 70% w/w, most preferably 30-60% w/w, within the composition.

[0020] The device to be filled may be any munition or subsystem, typically this may be a primer cup, initiator, or part of the explosive train.

[0021] The metal (M) oxide and a metal (M'), may each have an average particle size of less than 1000 nm, preferably less than 500 nm, more preferably less than 100 nm, yet more preferably less than 50 nm.

[0022] The metal oxide (M) may be any oxide of a metal, preferably the metal is selected from a transition metal, Al, In, Sn, Mg, Be, B, and Si or a mixture thereof. Preferably the transition metal may be selected from Sc, Bi, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Mo or Zr.

[0023] The metal (M') may be selected from a transition metal, Al, In, Sn, Mg, Be, B, Si or a mixture thereof, provided that the metal and the metal ion in the metal oxide are not selected from the same metal, ie (M)≠(M'). The transition metal for the (M') may be selected from Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu or Zn.

[0024] Particularly preferred mixtures may be bismuth (III) oxide/aluminium, iron (III) oxide/aluminium, and copper (II) oxide/aluminium, Molybdenum (VI) oxide/aluminium and manganese (III) oxide/aluminium.

[0025] The charging reagent may be selected from any reagent that is capable of forming a stable complex such as a chelate, ligand with the metal M' and metal (M) oxide such that they repel each other in said suspension. The repulsion may be caused by large steric hindrances and/or ionic repulsion. The charging reagent may be monodentate, bidentate ligand oligomeric, polymeric, ionic, or polar organic compounds, such as halo, polyethylenimine, phosphate ester, poly(dimethyldiallylammonium chloride), trioxadecanoic acid and acrylate-acrylamide copolymers.

[0026] In a highly preferred arrangement the suspension may comprise a non-ionic surfactant. The non-ionic surfactant may assist to prevent aggregated particles from forming as well as enhancing adsorption on the particle.

[0027] The selection of the non-ionic surfactant may prepare the surface of the metal (M') and metal (M) oxide particles to accept the charging reagent ligand, such as to facilitate the adsorption of the charging ligand, ie protons, iodide, conjugate systems, or ions on the particles surface.

[0028] A preferred example of a charging reagent may be iodine, acetone and deionized water in a polar organic solvent, with a non-ionic surfactant, such as acetylacetone or ethylene glycol.

[0029] The solvent is preferably an organic solvent, preferably a polar solvent, such as for example an C1 to C10 alcohol, such as alkyl, alkenyl, aromatic, straight or branch chained, and mixtures thereof.

[0030] The use of large complex ligands, may provide steric repulsion, electronic repulsion. The surfactant may itself form a chelate or complex bonds with the metal (M') and metal atom of the metal (M) oxide. Examples of the non-ionic surfactants may be acetylacetone, ethylene glycol or similar compounds.

[0031] In one arrangement the device may be initiated by an electrical stimulus, such as an electric spark, potential difference, which requires the use of a conducting composition. The nanothermite suspension may be made conducting by the inclusion of conducting particles, such that the composition may comprises a graphitic filler in the range of 1-40% by weight, a graphitic filler greater than 20%, such as in the range of 20 to 40% w/w. The graphitic filler had been found to further assist the complexing of the charging reagent and non-ionic surfactant with the metal and metal oxides.

[0032] The Resonant Acoustic Mixing stimulus process may be affected at different frequencies/power to provide homogeneous mixing of the composition. Resonant acoustic mixing is far removed from sonification (or ultrasound) techniques. Ultrasound employs very high frequencies, typically greater than 20 KHz.

[0033] In a highly preferred arrangement the resonant acoustic mixing may be caused at a frequency in the range of less than 200 Hz, preferably less than 100 Hz, preferably from 20 Hz to 100 Hz, more preferably in the range of from 50 Hz to 70 Hz, yet more preferably 58 Hz to 60 hz. The resonant acoustic mixing occurs at very low frequencies, in the order of tens of hertz, compared to those used in sonification (ultrasound), which is tens of thousands of hertz. Typically the resonant acoustic mixing stimulus may apply an acceleration force of up to 100 g.

[0034] Resonant acoustic mixing induces microscale turbulence by propagating acoustic waves of a low frequency throughout a mixture. The resonant acoustic mixing system has a lower frequency of acoustic energy and can be more

readily applied to larger scale of mixing than ultrasonic agitation. The mixing time for typical shear force mixers may be in the order of several hours to ensure homogenous mixing, in resonant acoustic mixing the stimulus may cause the time to be reduced to less than hour, more preferably less than 20 mins or even less than 5 minutes. The use of conventional mixing techniques with nanometric scale powders, particularly those in suspension, often causes high viscosity suspensions, the use of Resonant Acoustic Mixing is particularly suited to mixing high viscosity suspensions.

[0035] According to a further aspect of the invention there is provided a deposited nanothermite energetic material comprising a

[0036] i) a binder,

[0037] ii) a metal (M) oxide and a metal (M'), wherein the average particle size of the metal (M) oxide and a metal (M') is less than 100 nm,

[0038] wherein said metal (M) and metal (M') are complexed with a reagent such that said metal (M) oxide in said metal (M) oxide complex, and the metal (M') in the metal (M') complex have the same electrostatic charge.

[0039] According to a further aspect of the invention there is provided a method of filling explosive devices with a nanothermite initiatory composition, comprising the steps of:

[0040] forming a nanothermite composition comprising

[0041] (i) a binder,

[0042] (ii) a thermite suspension of a metal (M) oxide and a metal (M') in a solvent, wherein the average particle size of the metal (M) oxide and a metal (M') is less than 1000 nm, provided that $M \neq M'$,

[0043] (iii) forming a charging reagent comprising a reagent capable of forming a stable complex with each of the metal (M) oxide and the metal (M'), such that said metal (M) in said metal (M) oxide complex, and the metal (M') in the metal (M') complex have the same electrostatic charge or are sterically hindered, such that said metal (M) oxide complex and a metal (M') complex repel each other,

[0044] forming an admixture of the binder, thermite suspension and charging reagent, causing said admixture to be mixed under Resonant Acoustic Mixing to provide a stable suspension of an admixture of a nanothermite complex,

[0045] (iii) filling the device with the admixture of nanothermite complex.

Experimental

TABLE 1

Component (NTS_1)	Mass (g)	% (W/W)
Nanoparticle suspension:		
Nanothermite (n-Al/Bi ₂ O ₃)	5.887	39.249
Ethanol	5.414	36.093
Acetylacetone	2.291	15.276
"Charging reagent":		
Iodine	0.165	1.099
Acetone	0.233	1.550

TABLE 1-continued

Component (NTS_1)	Mass (g)	% (W/W)
Deionised water	0.147	0.981
Ethanol	0.863	5.752
Total:	15.000	100.00

[0046] The above suspension was prepared with a nanothermite loading of 40% w/w. The suspension was substantially devoid of aggregated particulates, the suspension was capable of being deposited onto a substrate and dried.

1. A method of filling an explosive device with a nanothermite initiatory composition, the method comprising:
- forming a nanothermite suspension of a metal (M) oxide and a metal (M') in a solvent, wherein an average particle size of the metal (M) oxide and the metal (M') is less than 1000 nm, provided that (M)≠(M');
 - forming a charging reagent comprising a reagent capable of forming a stable complex with each of the metal (M) oxide and the metal (M'), wherein the metal (M) and the metal (M') have a same electrostatic charge or are sterically hindered, such that the metal (M) oxide complex and the metal (M') complex repel each other;
 - forming an admixture of a binder, the nanothermite suspension, and the charging reagent;
 - causing the admixture of the binder, the nanothermite suspension, and the charging reagent to be mixed under Resonant Acoustic Mixing to provide a stable suspension of an admixture of a nanothermite complex; and
 - filling the explosive device with the admixture of the nanothermite complex.
2. The method according to claim 1, wherein the Resonant Acoustic Mixing is caused at a frequency in a range of less than 200 Hz.
3. The method according to claim 2, wherein the Resonant Acoustic Mixing is caused at a frequency in a range of less than 100 Hz.
4. The method according to claim 3, wherein the Resonant Acoustic Mixing is caused at a frequency in a range of 58 Hz to 60 hz.
5. The method according to claim 1, wherein a Resonant Acoustic Mixing stimulus applies an acceleration force of up to 100 g.
6. The method according to claim 1, wherein the nanothermite suspension comprises a binder, in the range of from 0.1 to 16% w/w.
7. The method according to claim 1, wherein the nanothermite suspension comprises a non-ionic surfactant.
8. The method according to claim 1, wherein the composition is deposited in a primer cup, initiator, or explosive train.

9. The method according to claim 1, wherein the metal (M) oxide is an oxide of a metal selected from a transition metal, Al, In, Sn, Mg, Be, B, and Si or a mixture thereof.
10. The method according to claim 9, wherein the transition metal is Sc, Bi, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Mo or Zn.
11. The method according to claim 1, wherein the metal (M') is selected from a transition metal, Al, In, Sn, Mg, Be, B, Si or a mixture thereof.
12. The method according to claim 11, wherein the transition metal is Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu or Zn.
13. The method according to claim 1, wherein the composition comprises a graphitic filler in the range of 20-40% by weight.
14. The method according to claim 1, wherein the solvent is a polar organic solvent.
15. The method according to claim 1, further comprising causing the nanothermite suspension to be evaporated, to provide a powered nanothermite encapsulated in the binder.
16. The method according to claim 1, wherein filling the explosive device includes applying the admixture of the nanothermite complex to the explosive device by extrusion, deposition, spray process, nozzle printing, or vapor deposition.
17. A method of forming a nanothermite initiatory composition, the method comprising:
- forming a nanothermite suspension of a metal (M) oxide and a metal (M') in a solvent, wherein an average particle size of the metal (M) oxide and the metal (M') is less than 1000 nm, provided that (M)≠(M');
 - forming a charging reagent comprising a reagent capable of forming a stable complex with each of the metal (M) oxide and the metal (M'), wherein the metal (M) and the metal (M') have a same electrostatic charge or are sterically hindered, such that the metal (M) oxide complex and the metal (M') complex repel each other;
 - forming an admixture of a binder, the nanothermite suspension, and the charging reagent; and
 - causing the admixture of the binder, the nanothermite suspension, and the charging reagent to be mixed under Resonant Acoustic Mixing to provide a stable suspension of an admixture of a nanothermite complex.
18. The method according to claim 17, wherein the Resonant Acoustic Mixing is caused at a frequency in a range of less than 200 Hz.
19. The method according to claim 17, wherein a Resonant Acoustic Mixing stimulus applies an acceleration force of up to 100 g.
20. A nanothermite initiatory composition formed according to the method of claim 17.

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